VAY

OPen Book!
No ChatGPT please.
Thanks.

→ Your exam has <u>11</u> questions.

- → Questions are numbered in red
- → The sections of any question are denoted with letters also in red
- → ~~Under~~ Below each question is a black circled number. This is the points for the respective question.

→ Your exam paper has <u>4</u> pages. Page numbers are in top-right corner of each page in blue circles.

→ You have 40 minutes to solve all the questions. The answers should be uploaded to Github as issues. You have 10 minutes for submitting your answersheets. (upload pictures ☺)

example of answering :-

<u>1</u> , <u>8,a</u> , <u>7,d</u>

question number and letter, underlined.

GOOD LUCK!

-:)

(1) (5)

1. → Why ~~set theory~~ did we start with set theory? (3)

2. → Consider the infinite set of even integers (2)

$E = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$. write E in set-builder notation.

$$E = \{ \underline{\qquad\qquad\qquad} \}$$

3. → write the following in set-builder notation:

   a. → $\{2, 4, 8, 16, 32, 64, \ldots\}$ (1)

   b. → $\{\ldots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \ldots\}$ (1)

   c. → $\{3, 4, 5, 6, 7, 8\}$ (1)

4. → find the following cardinalities.

   a. → $|\{x \in \mathbb{Z} : |x| < 10\}|$ (2)

   b. → $|\{x \in \mathbb{R} : |x| < 10\}|$ (2)

   c. → $|\{x \in \mathbb{N} : 5x \leq 20\}|$ (1)

5 → Sketch the following set of points on the 2-D plane. (x,y)

a 3 → $\{(x,y) : x \in [0,1], y \in [1,2]\}$

b 1,5 → $\{(x,y) : x = 2, y \in [1,2]\}$

c 1,5 → $\{(x,y) : x \in [0,1], y = 2\}$

6 4 → We stated that $A \times \emptyset$ for any set A equals to $\emptyset$. Proof this statement.

7 → Sketch these cartesian products on x-y plane

a 3 → $[0,1] \times [1,2]$

b 3 → $\{2\} \times [1,2]$

c 3 → $[0,1] \times \{2\}$

8 True or False! $\langle \in \rangle$ element of
$\langle \subseteq \rangle$ subset of

3

a 1 $\rightarrow 4 \in \{2,3,4\}$

b 1 $\rightarrow 7 \in \{2,3,4\}$

c 1 $\rightarrow 2 \notin \{2,3,4\}$

d 1 $\rightarrow \emptyset \in \{2,3,4\}$

e 1,5 $\rightarrow \emptyset \in \emptyset$

f 1,5 $\rightarrow N \in \{N\}$

g 1 $\rightarrow \emptyset \subseteq \{2,3,4\}$

h 3 $\rightarrow \emptyset \subseteq \emptyset$

i 2 $\rightarrow N \subseteq \{N\}$

j 4 $\rightarrow \{(1,2),(2,2),(7,1)\} \subseteq N \times N$

(4) → Power set of any set is the set of all subsets of that set. It's denoted as $P(A)$ for set A. (4)

Write the power sets of the following sets.

(a) → $\{1,2\}$ (2)

(b) → $\emptyset$ (3)

(c) → $\{\emptyset\}$ (2)

(d) → $\{\{0,1\},\{0,1\}\}$ (3)

(10) → Explain $P(\mathbb{R}^2)$! (6)

(11) → Show the following on the number line.

(a) → $[2,5] \cup [3,6]$ (3)

(b) → $[2,5] \cap [3,6]$ (3)

(c) → $[2,5] - [3,6]$ (3)

(d) → $[3,6] - [2,5]$ (3)

Extra Mile!

→ We stated the ~~the~~ complement of any set A is the set of all elements from the universe for A excluding elements of A. That is

$$\boxed{\bar{A} = U - A}$$

→ It's said that

$$\boxed{\overline{A \cup B} = \bar{A} \cap \bar{B}}$$

(complement of A union B equals to the intersection of their complements)

holds true.

Proof this claim.

→ This is requested for the next week.

→ Don't use chat GPT!!!

→ Good luck!